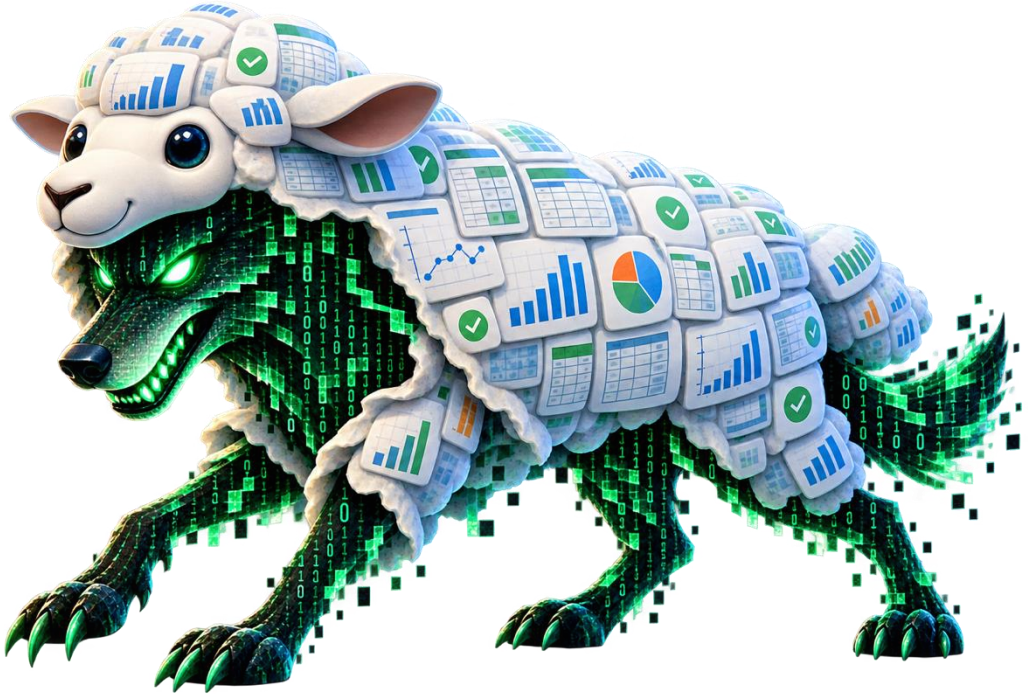


Is Your Data Lying?

Bayesian Data Credibility



Vince Nelidov

Staff Data Scientist @ Blue Yonder GmbH

Real life use case: Lead in drinking water

The problem

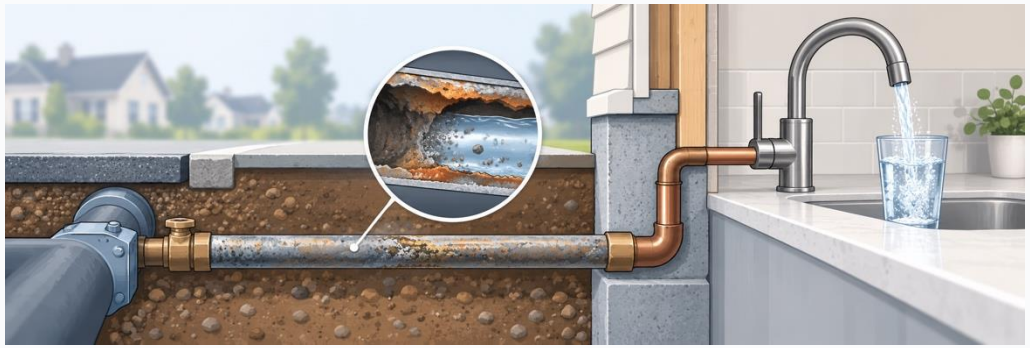
A city wants to know whether elevated lead is rare enough to avoid further investigation / replacement

The evidence

300 household tap-water measurements (self-reported)

The decision

Estimate citywide prevalence + the chance it exceeds 10% (policy violation)

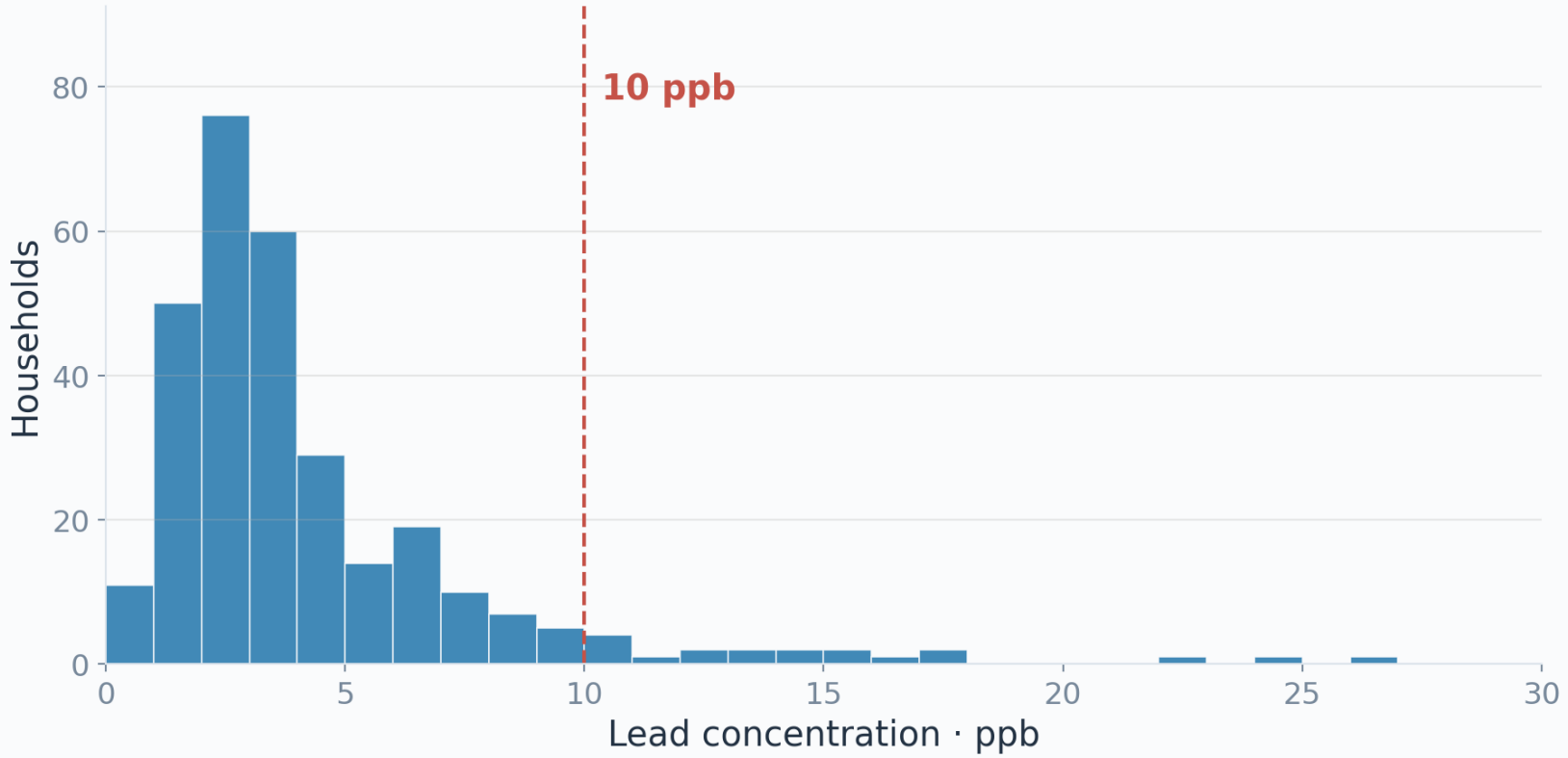


Policy reference Lead concentration must not exceed **10 ppb** level at the 90th percentile

Simulated data inspired by the 2015 Flint water surveys · TidyTuesday 2025-11-04

Lead concentrations in the original 300 samples

At first glance, the overall distribution looks reassuring



Original survey

300

households

Above 10 ppb

19 / 300

6.3% of samples

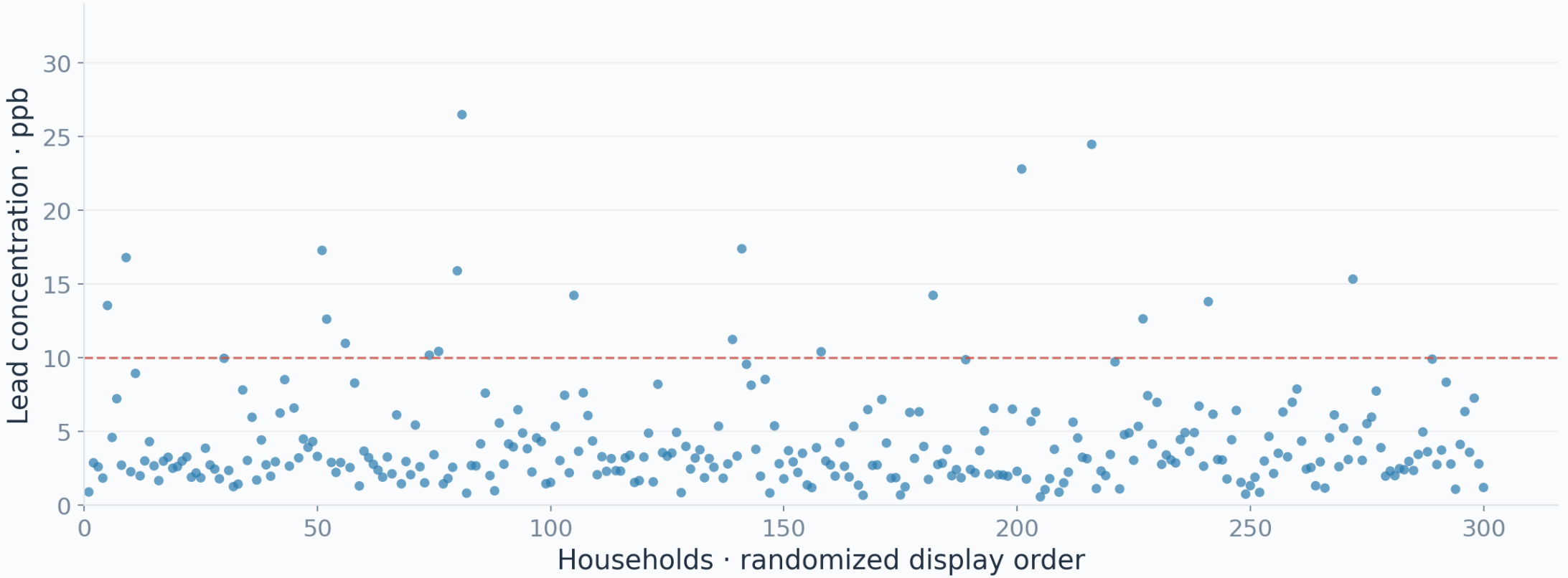
90th percentile

8.2 ppb

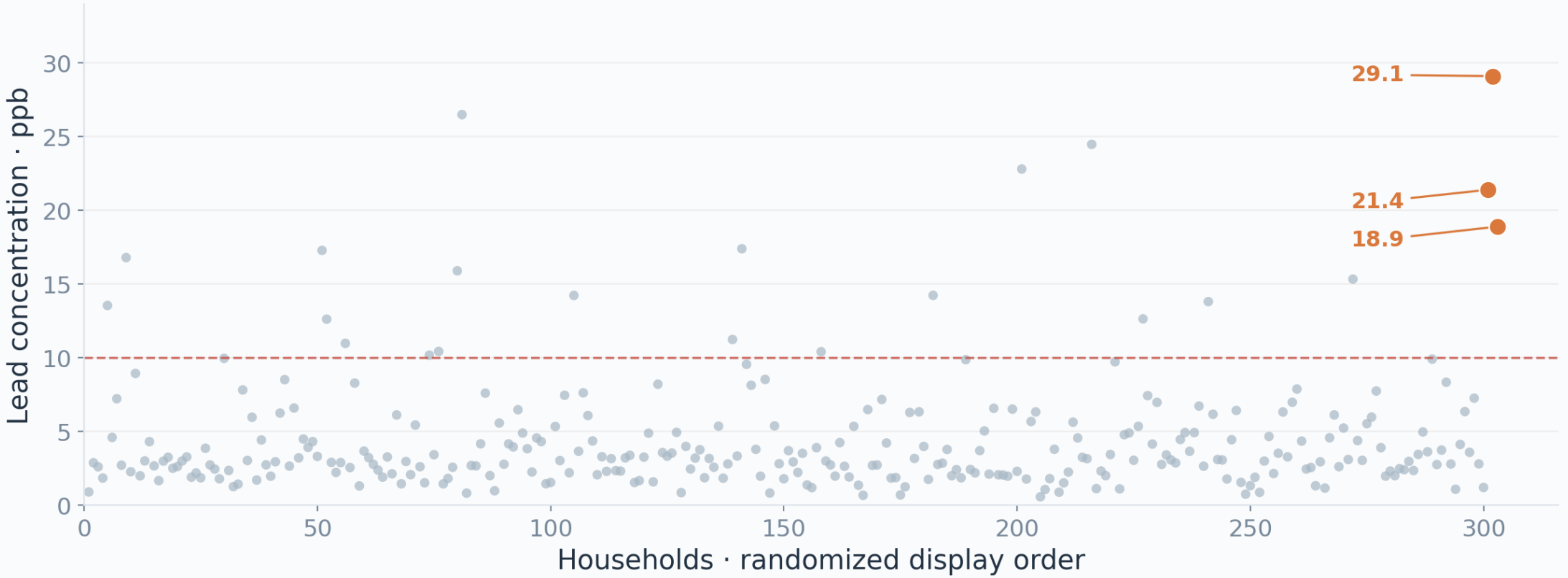
observed

Can we trust this data and conclude the investigation?

Let’s have a closer look



Three more households submit results after the deadline



Are these new values surprising? How likely are we to randomly observe this? → Bayesian Inference!

Bayesian inference: what changes after we see data?

Combining prior knowledge with new evidence to quantify uncertainty

Posterior ∝ Prior × Likelihood

PRIOR $p(\theta)$

Before measurement

Which concentration distributions do we think are plausible?

(like prior distributions of μ & σ)

LIKELIHOOD $p(D \mid \theta)$

What the sample tells us

Which distribution parameters better explain observations?

(e.g. lognormal model)

POSTERIOR $p(\theta \mid D)$

After measurement

Which distributions are plausible now, and how uncertain are we?

(μ & σ plausible given data)

From posterior draws, we can...

Estimate *risk* of policy breach for each plausible parameter combination

Predict new measurements and associated confidence

Propagate uncertainty instead of single point estimates

Fitting a Bayesian lognormal model in Python

The prior, likelihood and posterior + sampling -- now in Python with PyMC library

```
import numpy as np
import pandas as pd
import  pymc as pm
from scipy.stats import norm

y = pd.read_csv("water.csv") ["lead_ppb"].values

with pm.Model() as model:
    v = pm.InverseGamma("v", alpha=2.25, beta=0.45)
    mu = pm.Normal("mu", mu=np.log(3.4),
                   sigma=pm.math.sqrt(v / 0.25))
    sigma = pm.Deterministic("sigma", pm.math.sqrt(v))

    pm.LogNormal("lead", mu=mu, sigma=sigma,
                 observed=y)

    idata = pm.sample(2000, tune=1000, chains=4,
                      random_seed=42)

posterior = idata.posterior
p = norm.sf(np.log(10), posterior["mu"].values,
            posterior["sigma"].values)
```

What we fit

A distribution of possible lead concentrations.

What we sample

Many plausible values for μ and σ.

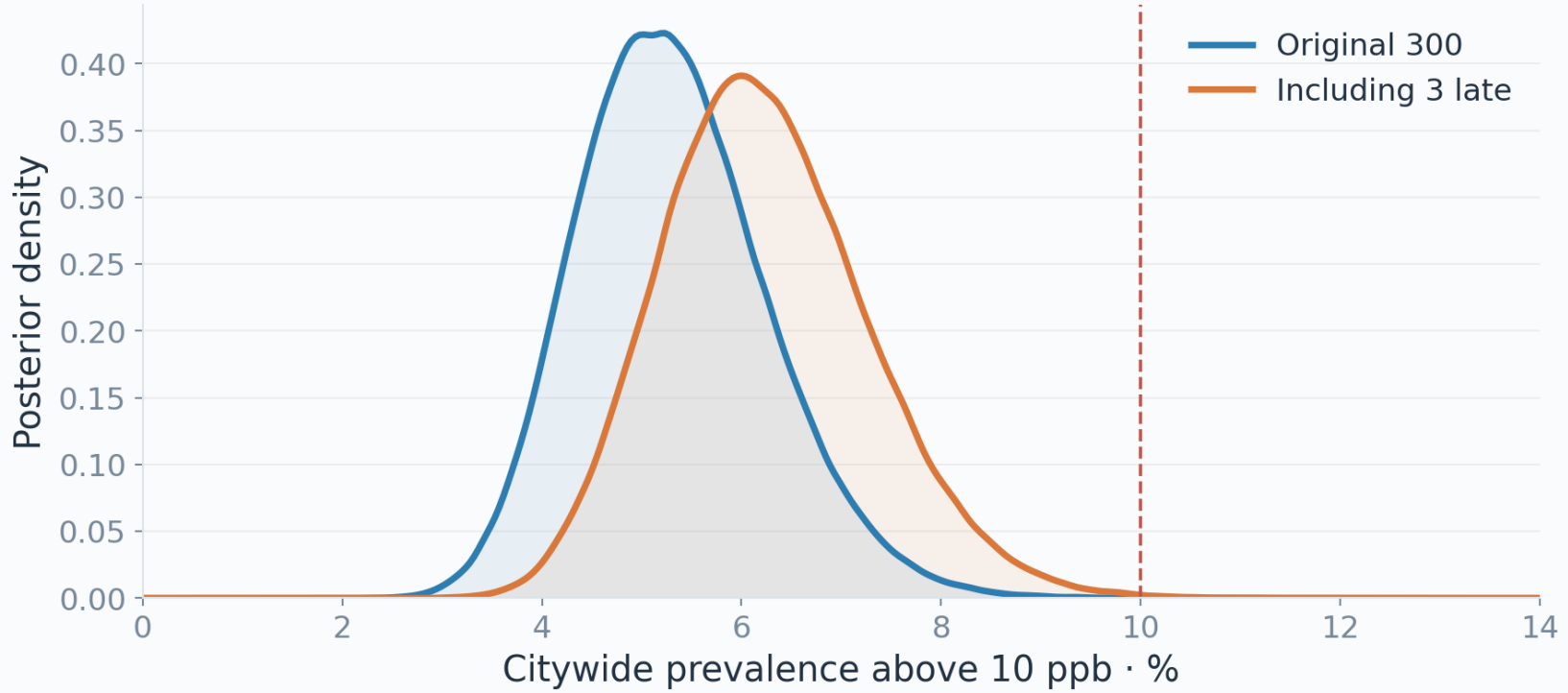
What we derive

One exceedance probability per posterior draw.

See also: Stan / NumPyro · ArviZ for diagnostics

Result: the posteriors and how the new data changes it

Prevalence = share of homes above 10 ppb · Risk of breach = Pr(prevalence > 10%).



Original 300

5.2%

95% interval 3.7–7.3%

After 3 late

6.2%

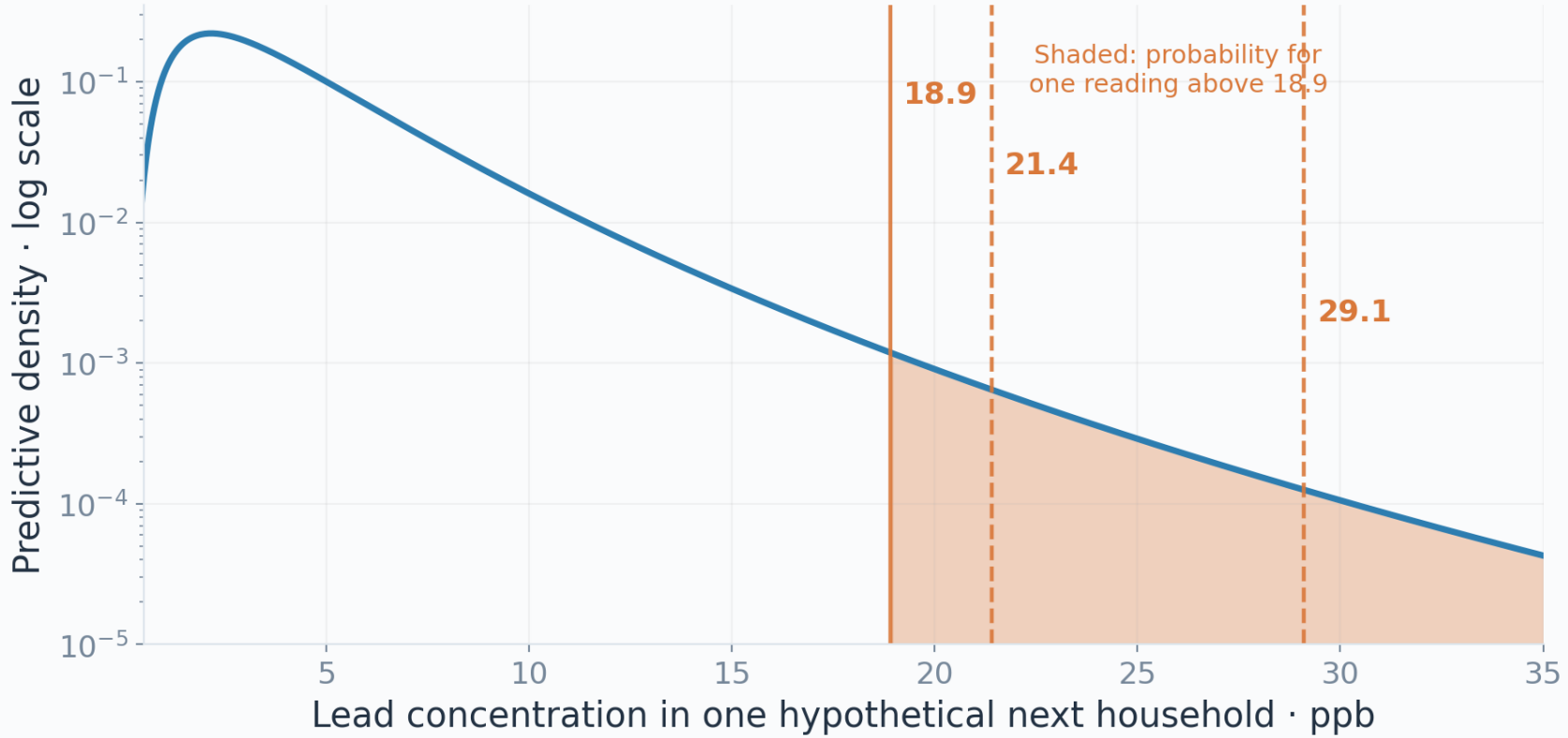
95% interval 4.4–8.4%

Risk of breach

< 0.1%

Could our original model have produced this triple?

Posterior predictive distribution for one new household



All 3 ≥18.9 ppb

≈ 1 in 5 million

Very surprising together

A single high reading is possible. Three this elevated are unlikely under the pooled model.

Note: late responses may not be random draws

Three ways the same dataset can mislead us

01

WHO IS COVERED

Sampling bias

Known mismatch in housing composition vs reality

02

HOW WAS IT MEASURED

Unreliable measurements

Measurement techniques differ between observations

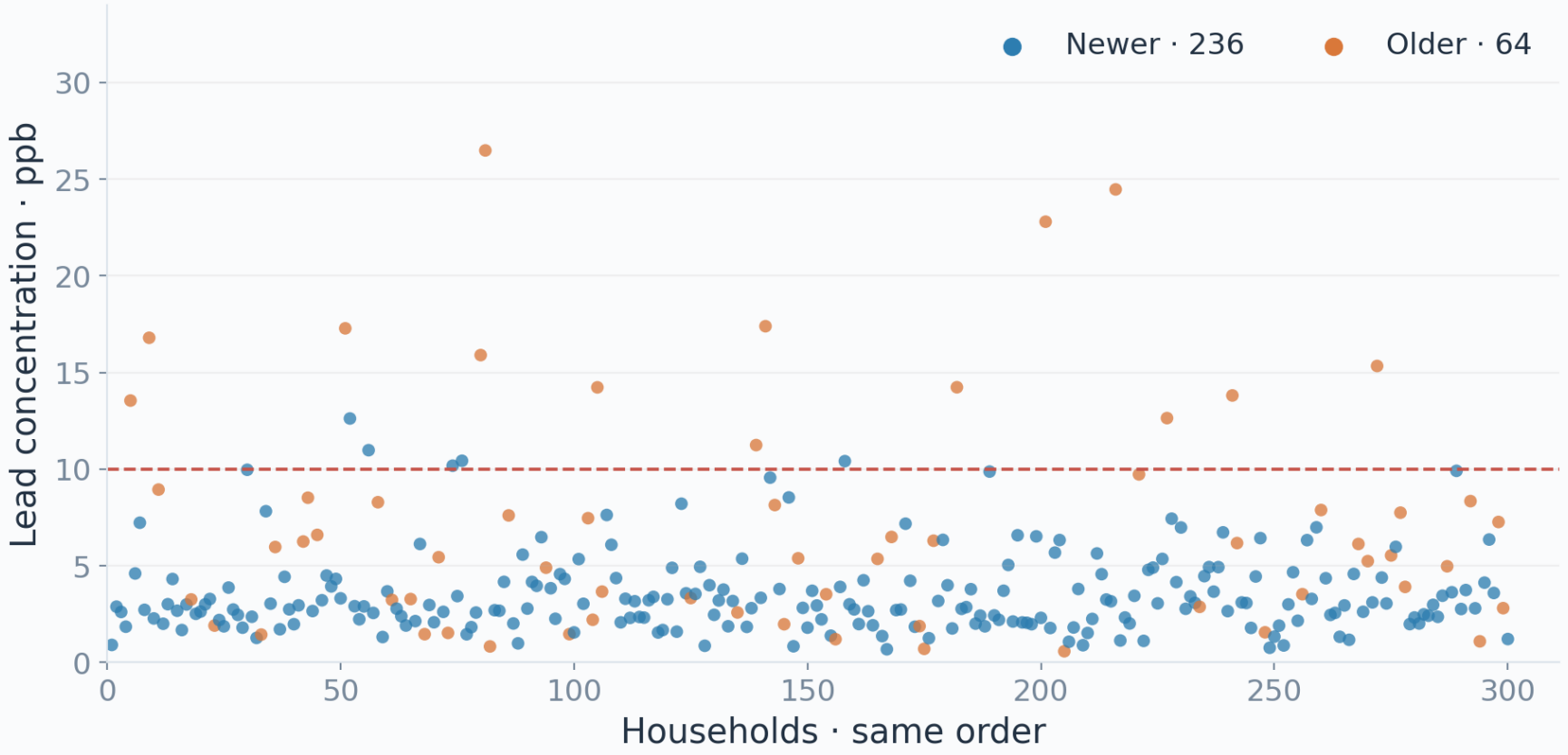
03

WHO IS MISSING

Hidden selection bias

Unobserved participation linked to outcomes

1 · Sampling bias - which homes did we survey?



Older in sample

21%

64 / 300

Older in city

45%

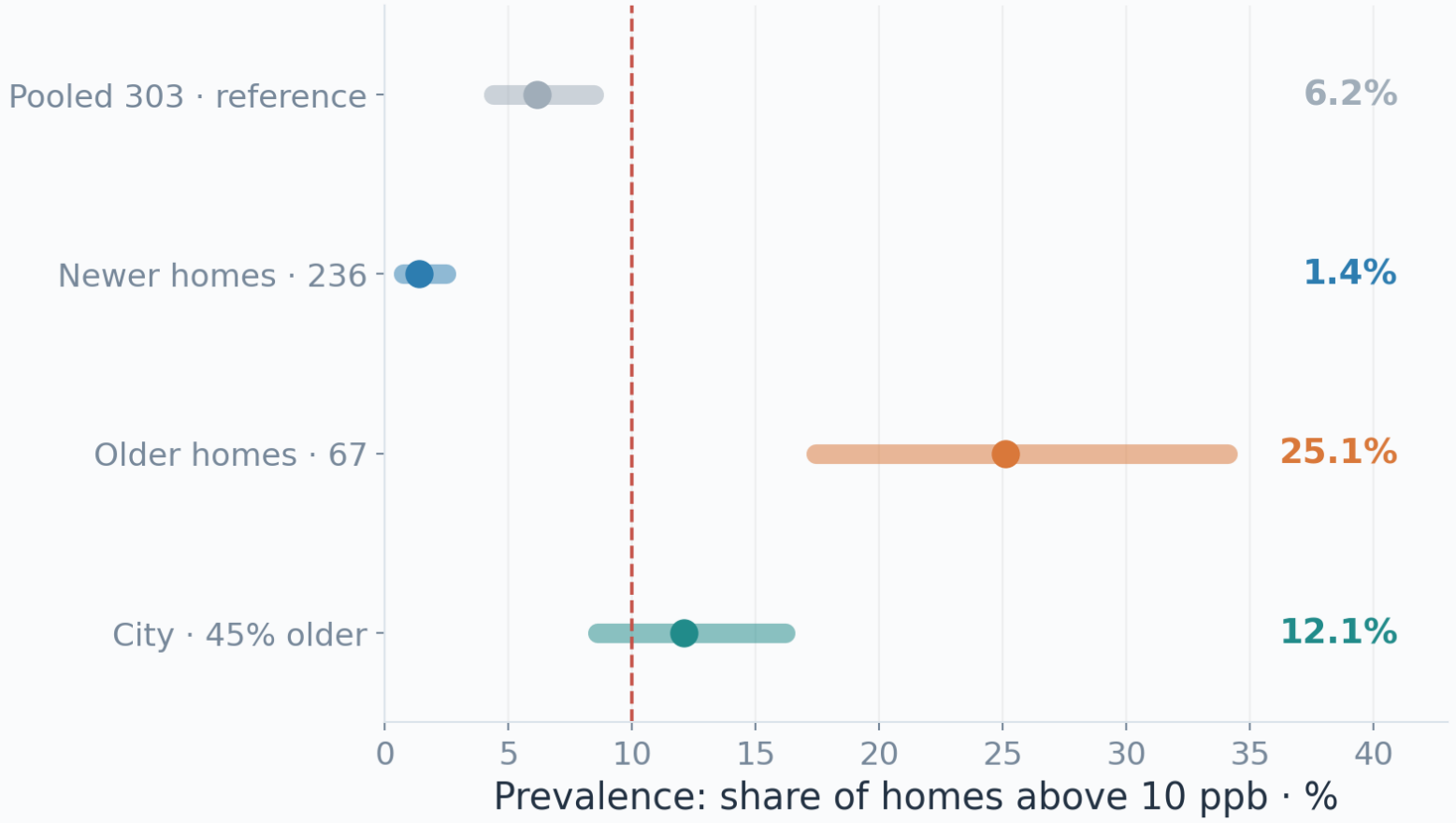
municipal housing register

14 of 19 original exceedances: older homes

All 3 late submissions: older homes

1 · Sampling bias - poststratification

The grey pooled estimate is our reference; orange and blue are separately fitted groups.



Split

236 newer · 67 older (3 late included)

Population weights

Municipal housing register: 55% newer / 45% older

```
p_city=.55*p_new+.45*p_old
(p_city > .10).mean()
```

City prevalence 12.1% · Breach risk 87%

For advanced follow up, see also: *Bayesian hierarchical models*

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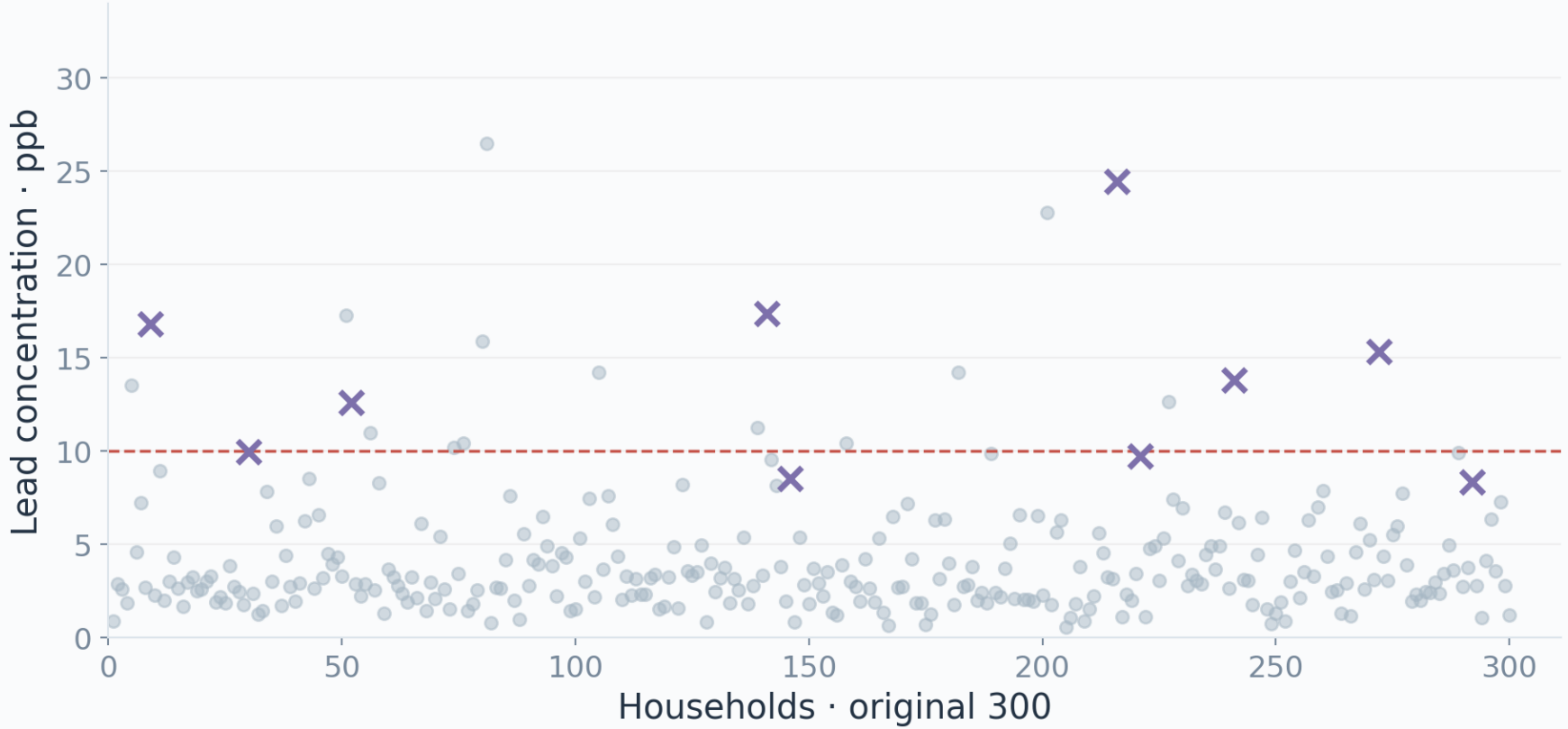
03

WHO IS MISSING

Hidden selection bias

Unobserved participation linked to outcomes

2 · Unreliable measurements - ten flagged data points example



Flagged tests

10 / 303

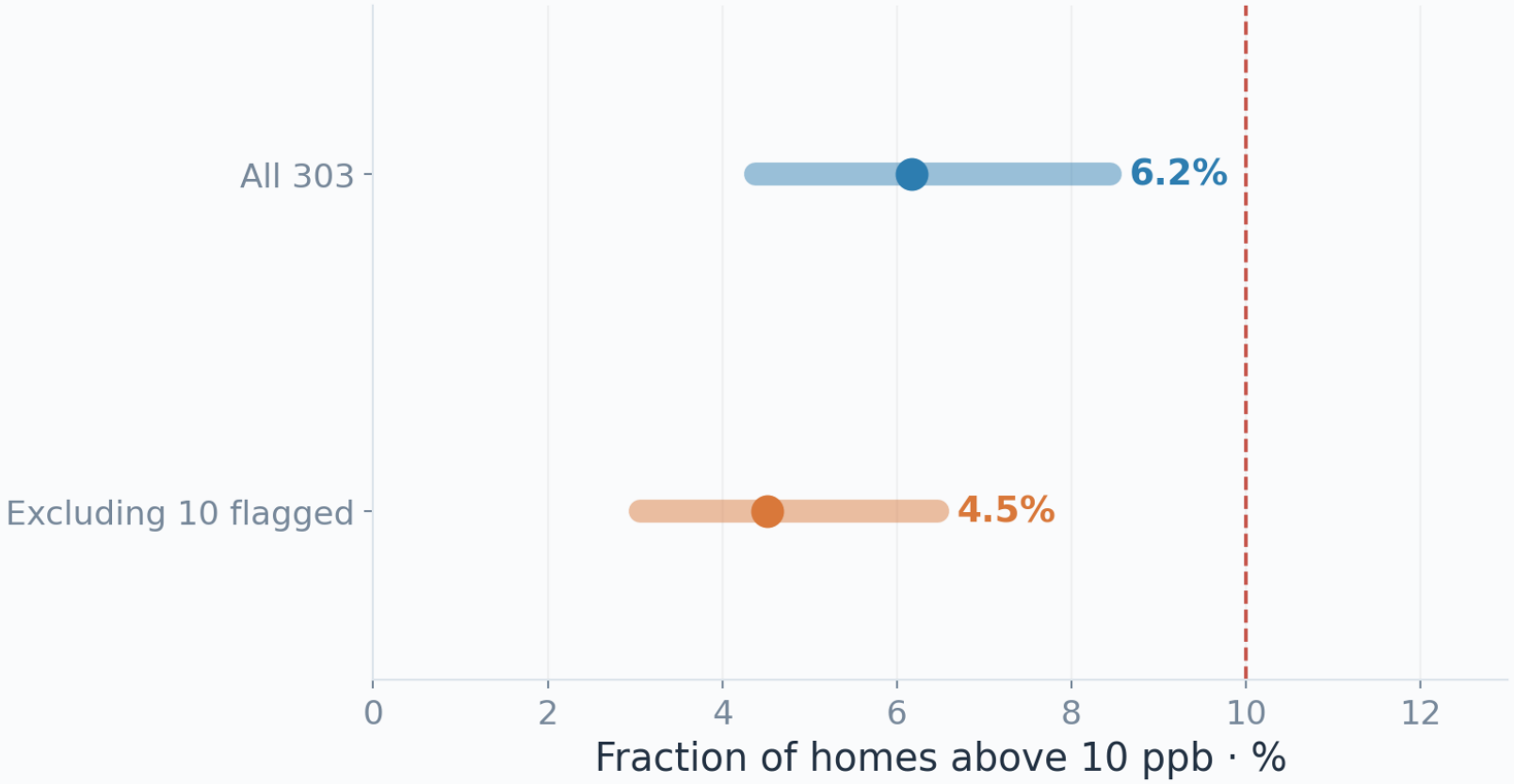
hot-tap sampling

Flags above 10 ppb

6 / 10

Cold first-draw protocol was intended. The recorded high readings may not be comparable.

2 · Unreliable measurements — how sensitive is our estimate?



All 303

6.2%

posterior median

Exclude 10 flags

4.5%

posterior median

Remove 10 at random instead:

6.2% median

Three ways the same dataset can mislead us

01

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Sampling bias

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Measurement techniques differ between observations

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WHO IS MISSING

Hidden selection bias

Unobserved participation linked to outcomes

3 · Hidden selection bias - what if respondents differ?

Only 300 households replied out of 1000 survey invitations

λ = relative response rate of high-lead homes versus other homes

ASSUMPTION		PREVALENCE	95% CREDIBLE INTERVAL	RISK OF BREACH
λ = 1.00	Equal chance of replying	6.2%	4.4–8.4%	< 0.1%
λ = 0.65	High-lead homes 35% less likely to reply	9.2%	6.6–12.4%	29.7%
λ = 1.50	High-lead homes 50% more likely to reply	4.2%	3.0–5.8%	< 0.1%

Recalculate every posterior draw

```
p_adj = p / (λ + (1 - λ) * p)
```

Then take its 2.5% and 97.5% quantiles

```
np.quantile(p_adj, [0.025, 0.975])
```

Three assumptions, three answers

Each alternative starts from the same 303-row baseline - we do not add the adjustments together



Risk of breach

< 0.1%

87%

< 0.1%

30%

TLDR: Don't Just Estimate the Answer - Question the Evidence

Before trusting your dataset - check:

WHO

Population coverage

HOW

Measurement protocol

WHO IS MISSING

Selection into the data

Use Bayesian Inference

Quantify uncertainty · Test predictions · Challenge assumptions





Thank you for your attention!

Add me on for any questions